03 - First Text Models

Objective

The objective is to obtain the first simple modelling results for the Rakuten product-classification task. Two models are compared:

1. a minimum baseline that always predicts the largest class;
2. a text baseline using TF-IDF and a Linear SVM.

Both models use the existing shared training/validation split.

1. Load the Prepared Text

The processed file contains cleaned text, the correct product category and the shared split. Product and image identifiers are not used as model features.

```
Training products: 67,932
Validation products: 16,984
```

2. Largest-Class Baseline

This baseline predicts the most common training category for every validation product. It ignores the combined text completely and establishes a minimum score.

```
Largest class: 2583
Weighted F1: 0.0258
Macro F1: 0.0080
```

3. Convert Text into TF-IDF Features

This section converts the cleaned product text into numerical TF-IDF features. TF-IDF gives more importance to distinctive terms and less importance to words that occur across many products. The vocabulary is learned only from the training data to prevent data leakage. The vocabulary is learned from training text only. The fitted vectorizer then transforms the validation text without learning from it.

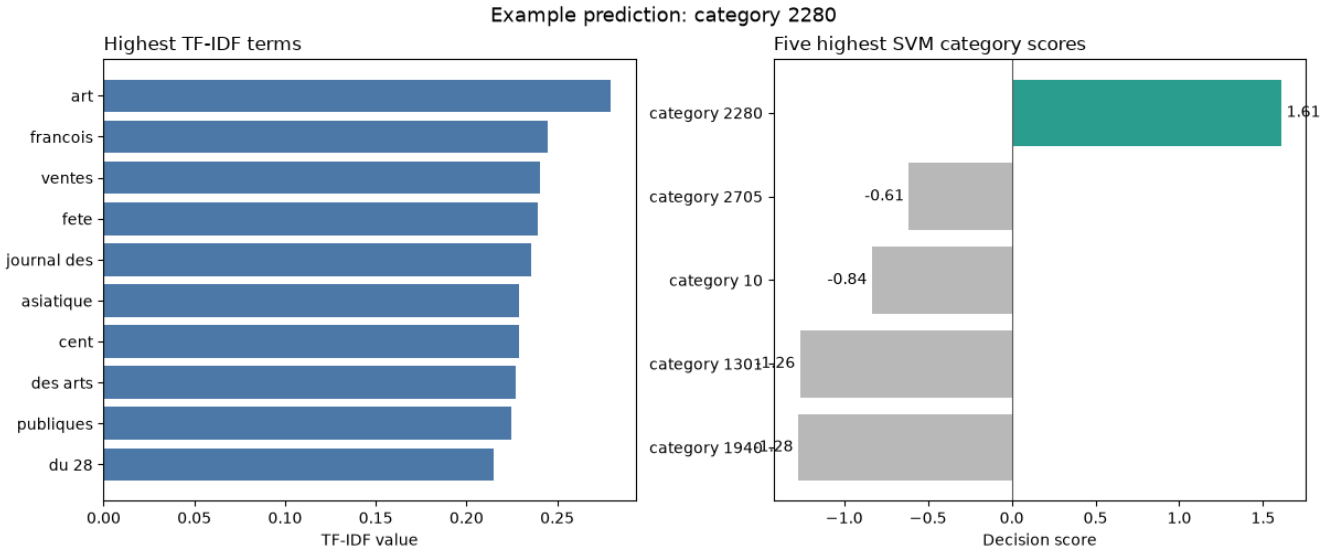
```
Training matrix: (67932, 50000)
Validation matrix: (16984, 50000)
```

4. Train the First Text Classifier

A Linear SVM is a standard, efficient baseline for high-dimensional TF-IDF text features. It learns from the training products and predicts categories for the unseen validation products.

Figure 1 — TF-IDF features and SVM scores

TF-IDF assigns importance to terms in a product. The SVM uses these features to calculate one score per category and selects the highest score.



Analysis

TF-IDF gives the largest values to the most distinctive terms in the product. The SVM combines these features and assigns the highest decision score to the predicted category.

5. Compare the Results

Weighted F1 gives more influence to large classes. Macro F1 gives equal importance to every class and therefore reveals whether smaller classes are being neglected.

	model	weighted_f1	macro_f1
0	Largest-class baseline	0.0258	0.0080
1	TF-IDF + Linear SVM	0.8274	0.8156

6. Did the Text Cleaning Help?

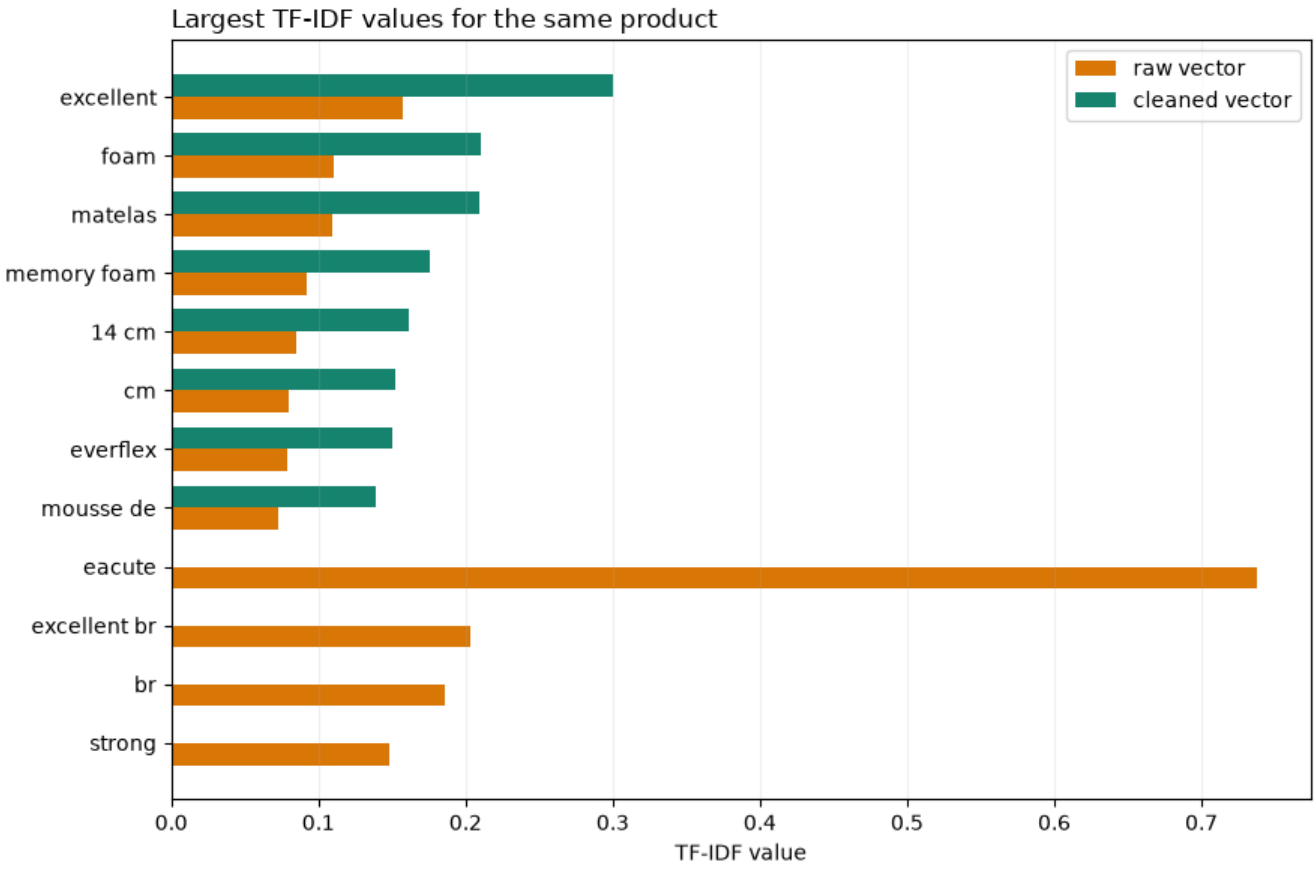
To measure the effect of cleaning, the same model is trained again on the raw text. Only the text changes: same products, same split, same TF-IDF settings and same Linear SVM. Any difference in score therefore comes from the cleaning.

The raw text is the original designation and description joined together, with nothing removed.

	text_raw	text_combined
1	Journal Des Arts (Le) N° 133 Du 28/09/2001 - L...	journal des arts (le) n° 133 du 28/09/2001 - l...
6	Christof E: Bildungsprozessen Auf Der Spur	christof e: bildungsprozessen auf der spur
12	Peaceable Kingdom Wheres Bear? The Hide And Fi...	peaceable kingdom wheres bear? the hide and fi...

Figure 2 — Raw and cleaned TF-IDF vectors

The same product is transformed by both vectorizers. Terms with one bar were removed or introduced by cleaning; terms with two bars remain in both versions.



Analysis

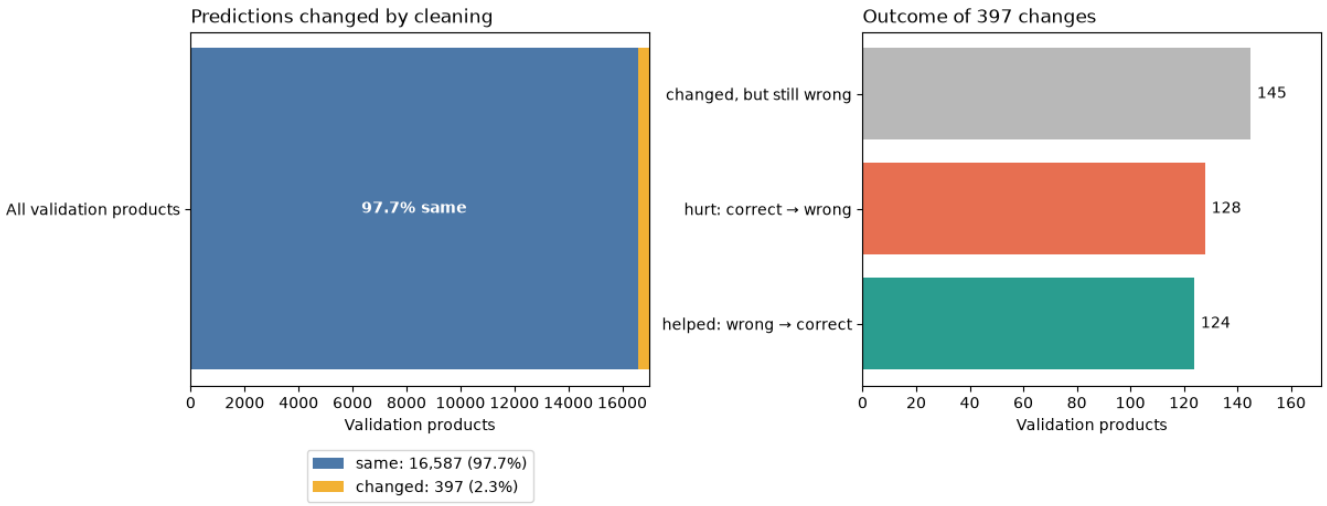
Cleaning removes some raw-text features and introduces corrected terms, but the main product words remain in both vectors. The representation therefore changes only partly, which helps explain why most SVM predictions remain unchanged.

	text used	weighted_f1	macro_f1
0	raw text	0.8276	0.8167
1	cleaned text	0.8274	0.8156

```
Validation products with HTML tags in the raw text: 18.7%
Products where both models predict the same class: 97.7%
```

Figure 3 — Effect of cleaning on predictions

The first plot counts changed predictions. The second shows whether those changes helped or harmed classification.




```
Cleaning changed 397 of 16,984 predictions (2.3%).
Net change in correctly classified products: −4.
```

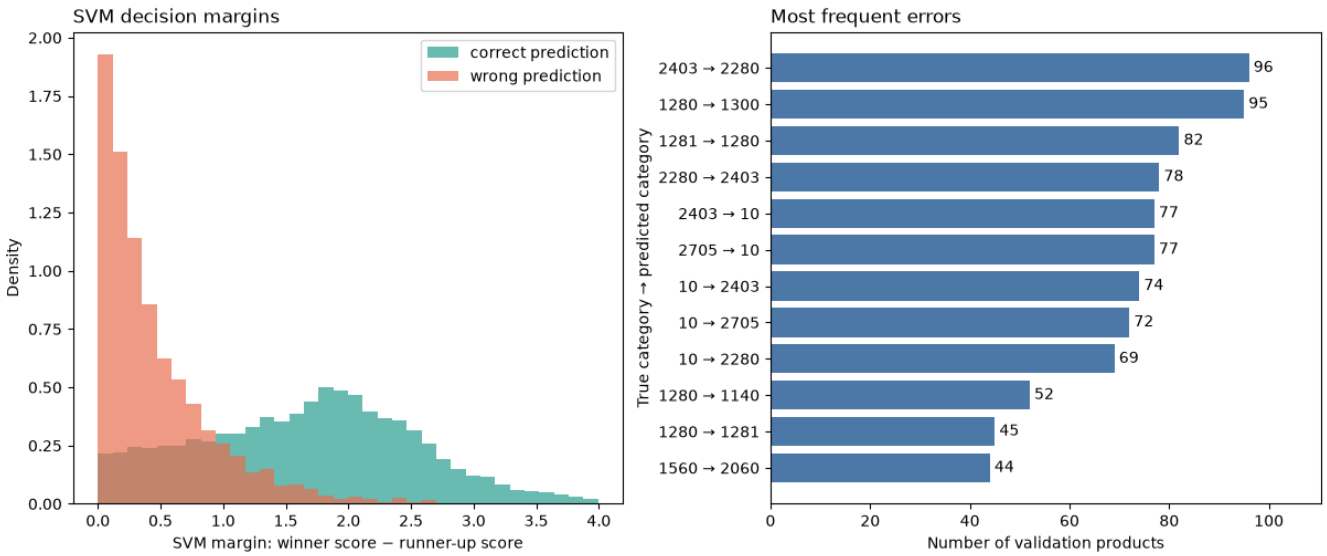
Analysis

Cleaning changed only **2.3%** of predictions and reduced the number of correct predictions by four. Most category information is already present in product titles, while TF-IDF can also learn repeated raw-text patterns. Cleaning is retained for readable features and later transformer models.

7. SVM Results

Figure 4 — Decision margins and frequent errors

The decision margin is the difference between the two highest category scores. Small margins indicate close decisions. The second plot shows the most frequent category confusions.



Analysis

Linear SVM is suitable for the sparse 50,000-feature TF-IDF matrix and provides a fast, strong baseline. Errors have smaller decision margins and concentrate in similar categories.

The next experiment should combine word and character TF-IDF, then tune the SVM parameter `C`. Character features can capture spelling variation, model numbers and encoding noise. Logistic Regression, transformers and text-image fusion can be compared afterwards.

8. Plot the TF-IDF Space and the SVM Errors

Figure 5 — Validation products in the TF-IDF space

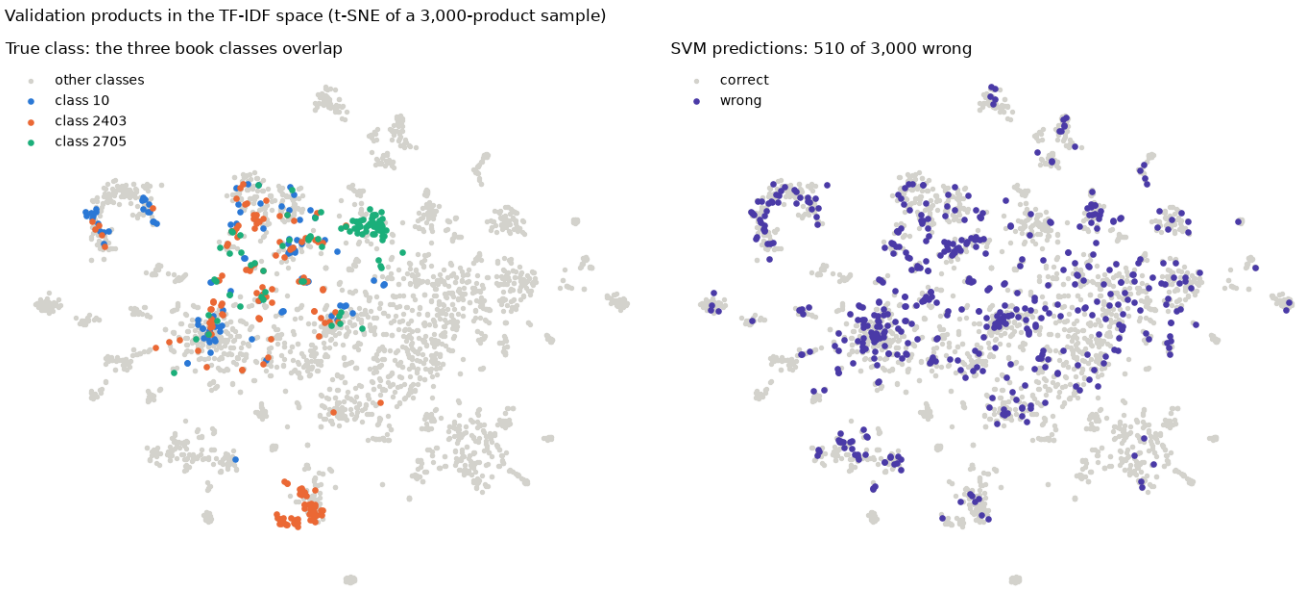
Each product is a point with 50,000 TF-IDF values, which cannot be drawn directly. Two steps reduce it to two dimensions:

1. **Truncated SVD** compresses the 50,000 values into 50. It is fitted on training rows only.
2. **t-SNE** places the products on a 2D map so that products with similar text sit close together.

A random sample of 3,000 validation products keeps t-SNE fast.

The map shows neighbourhoods, not the real SVM boundary. The SVM separates classes in all 50,000 dimensions, so its boundary cannot be drawn in 2D. Distances between far-apart groups on a t-SNE map have no meaning.

The left map highlights classes 10, 2403 and 2705. They are the three book classes and the most frequent confusions of the SVM. The right map shows which sampled products the SVM classifies wrongly.



Analysis

- Many classes form their own tight group. Products in these groups are almost always classified correctly.
- The three book classes are mixed together in the same region. Their titles use similar words, so TF-IDF places them close together.
- The SVM errors gather in these mixed regions and in the crowded centre of the map, where products from different classes share general words.
- This suggests that the remaining errors come from classes whose text is too similar, not from a bug. Images may help separate them later.

Conclusion

The largest-class baseline provides the minimum reference score. The TF-IDF model tests whether the cleaned product text contains useful category information. These are initial results, not a final optimized model.

The next text-modelling step is to examine per-class performance and the most frequent classification errors before changing model settings.